

INTRODUCTION TO STATISTICAL METHODS

Exponential Smoothing

Smoothing to forecast, the smoothing constant, and Holt / Holt–Winters for trend and seasonality.

Companion reader · Read beside the slides

How to use this companion. Colour boxes guide you: **blue** intuition, **amber** everyday picture, **green** worked example, **red** watch out, **purple** key takeaway. Read it next to the slides, in order.

1 Smoothing to forecast

A noisy series is hard to read. **Smoothing** averages recent values to reveal the underlying level and make a forecast. A plain **moving average** weights the last few points equally, but that ignores that *recent* data is usually more relevant. **Exponential smoothing** fixes this: it weights recent observations more heavily, with weights that *decay exponentially* into the past.

2 Simple exponential smoothing

The smoothed value S_t is a blend of the latest observation and the previous smoothed value:

$$S_t = \alpha x_t + (1 - \alpha) S_{t-1}, \quad 0 < \alpha < 1.$$

The **smoothing constant** α sets the memory: large α (near 1) reacts fast to new data (little smoothing); small α (near 0) is sluggish and very smooth. The one-step forecast is simply the latest smoothed value.

 Worked example: smoothing a short series ($\alpha = 0.3$)

Data $x = 10, 12, 13, \dots$, start $S_1 = x_1 = 10$.

$$S_2 = 0.3(12) + 0.7(10) = 3.6 + 7 = \mathbf{10.6},$$

$$S_3 = 0.3(13) + 0.7(10.6) = 3.9 + 7.42 = \mathbf{11.32}.$$

The forecast for period 4 is $S_3 = \mathbf{11.32}$. Unrolling the recursion shows the weights on past observations are 0.3, $0.3(0.7) = 0.21$, $0.3(0.7)^2 = 0.147, \dots$ — an exponential decay that always sums to 1.

3 Adding trend and seasonality

Simple exponential smoothing assumes a roughly *constant* level, so it lags behind a trend. **Holt's method** (double exponential smoothing) adds a second equation that smooths the *trend*, so forecasts follow a slope. **Holt–Winters** (triple exponential smoothing) adds a third component for **seasonality**, handling series with both a trend and a repeating pattern.

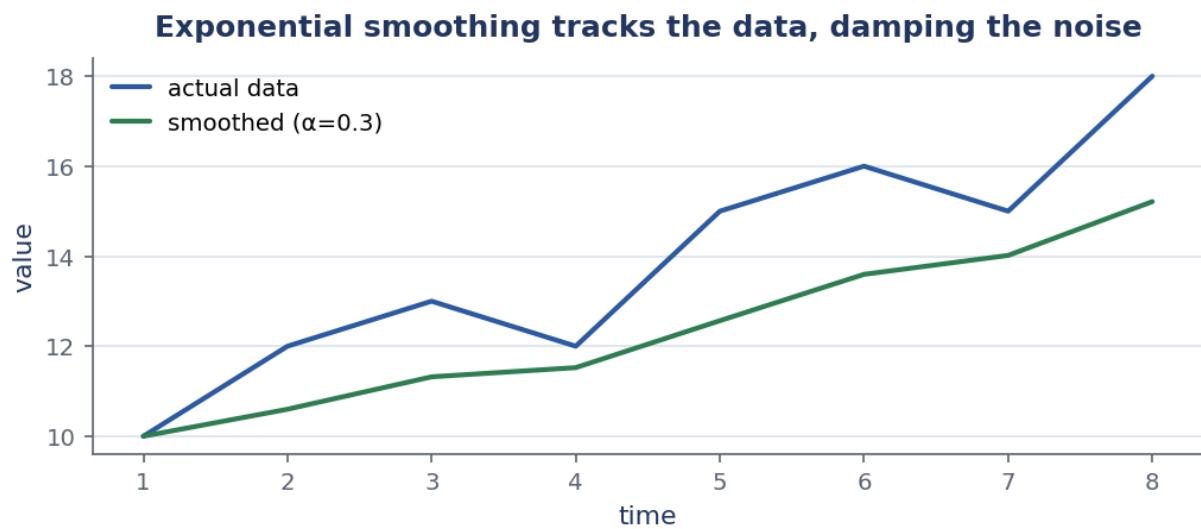


Figure 1: Exponential smoothing (here $\alpha = 0.3$) follows the data while damping the noise; a larger α hugs the data more tightly, a smaller one is smoother.

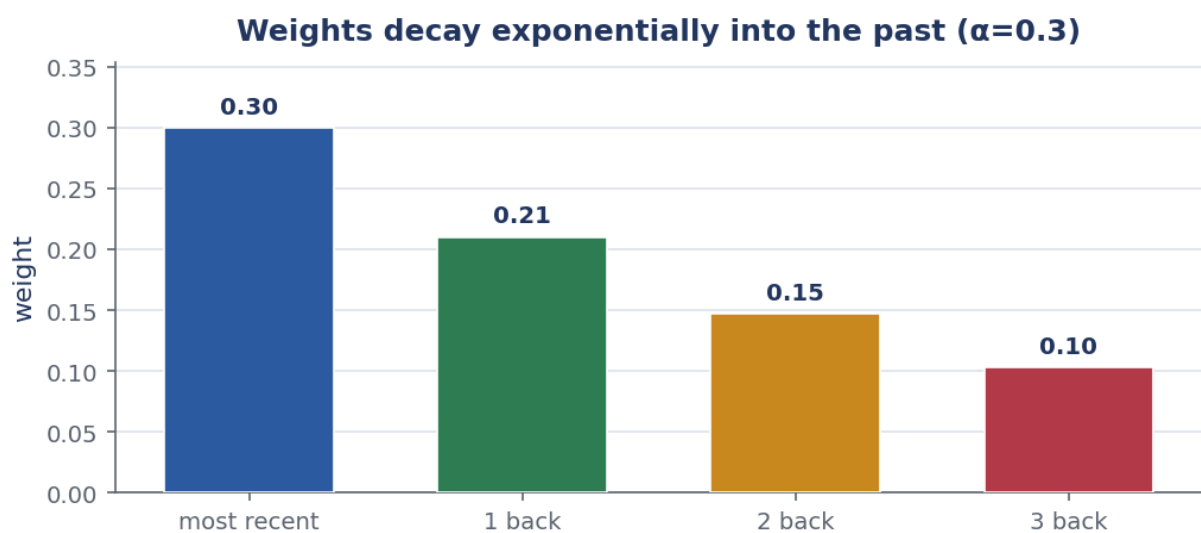


Figure 2: Each past observation's weight is $\alpha(1 - \alpha)^k$ — recent points count most, and the weights decay geometrically and sum to one.

✓ Key takeaway

Exponential smoothing forecasts with $S_t = \alpha x_t + (1 - \alpha)S_{t-1}$: recent data weighted most, weights decaying as $\alpha(1 - \alpha)^k$. α trades responsiveness (large) against smoothness (small); the one-step forecast is S_t (worked: $S_2 = 10.6$, $S_3 = 11.32$). **Holt** adds trend and **Holt–Winters** adds seasonality.